

WGS FINAL REPORT (SYNTHETIC)

Patient ID	SIM-DMD-01	Ordering Physician	Dr. [Ordering Physician]	Specimen	Simulated peripheral blood
Sex	Male	Account#	SIM-PIPE	Collected	2026-07-15
Date of Birth	XX/XX/XXXX (synthetic)	Source	Pipeline v1.1 test dataset	Received	2026-07-15
Indication	Simulated: suspected dystrophinopathy, elevated CK	Case ID	CS-DMD_INXXXX	Reported	2026-07-15

TEST RESULT: Two independent pathogenic findings in **DMD** were returned by this validation run: an in-frame exon 45–47 deletion, and a nonsense (stop-gained) single-nucleotide variant predicted to cause a frameshift/truncating effect. Pathogenic DMD variants cause **Duchenne/Becker Muscular Dystrophy** [MIM: 310200 / 300376], X-linked disorders of progressive muscle weakness whose severity depends heavily on whether the reading frame is preserved.

PATIENT PHENOTYPE (SIMULATED)

Simulated phenotype terms used to drive prioritization: progressive muscle weakness, calf pseudohypertrophy, elevated creatine kinase.

PRIMARY FINDINGS — Variants in genes associated with patient's simulated phenotype

Confirmation Status	Gene / Locus	Condition	Chrom : Coordinates	Variant	Zygosity	Inheritance	Classification
Confirmed	DMD	Becker Muscular Dystrophy (in-frame)	chrX: 31,950,000	Exon 45-47 deletion (in-frame)	Hemizygous	X-linked Recessive	Pathogenic
Confirmed	DMD	Duchenne Muscular Dystrophy (truncating)	chrX: 32,300,500	c.? p.? (stop_gained)	Hemizygous	X-linked Recessive	Pathogenic

INTERPRETATION

Variant Information

Note: this synthetic validation dataset bundles two independently generated DMD test variants in a single sample so that both the pipeline's SV/exon-CNV branch and its SNV branch are exercised in one run; the two would not necessarily co-occur in a single individual.

Exon 45–47 deletion: a large deletion spanning exons 45–47 is classified Pathogenic by ClassifyCNV and Likely Pathogenic by SvAnna. Because removing exons 45–47 together keeps the reading frame intact, this is predicted to produce a milder, Becker-like phenotype, consistent with the reading-frame rule used clinically to predict DMD/BMD severity. DMD is haploinsufficiency-sufficient per ClinGen (HI=3), supporting pathogenicity of exon-loss events.

Nonsense variant: a stop-gained SNV (CADD phred 34.0) is classified Pathogenic by both ClinVar (PVS1, PM2) and InterVar, and is predicted to cause a frameshift/truncating effect — the classically severe, Duchenne-like mechanism, in contrast to the in-frame deletion above. A common, benign synonymous variant was also detected in the DMD region and does not affect interpretation.

Gene Information

DMD encodes dystrophin, a structural protein that links the muscle cytoskeleton to the extracellular matrix and stabilizes the sarcolemma during contraction. Out-of-frame or truncating variants that abolish dystrophin expression cause the severe Duchenne phenotype; in-frame variants that permit expression of a shortened but partially functional protein typically cause the milder Becker phenotype — the basis of the reading-frame rule applied in the Interpretation above.

REFERENCES

Aartsma-Rus A, et al. Human Mutation. 2009. Entries in the Leiden Duchenne muscular dystrophy mutation database: an overview of mutation types and paradoxical cases that confirm the reading-frame rule.

Monaco AP, et al. Genomics. 1988. An explanation for the phenotypic differences between patients bearing partial deletions of the DMD locus.

Richards S, et al. Genetics in Medicine. 2015. Standards and guidelines for the interpretation of sequence variants (PMID: 25741868).

RECOMMENDATIONS

- Clinical correlation with the patient's phenotype is recommended.
- Because this dataset is synthetic pipeline-validation data, no clinical action should be taken on it.
- Genetic counseling is recommended in real cases to discuss implications for the patient and family.
- Orthogonal confirmation (e.g. Sanger sequencing, long-read sequencing, or inverse-PCR where applicable) is recommended before clinical sign-out of any pathogenic or likely-pathogenic call.
- Periodic re-evaluation of variant classification is appropriate as knowledge and databases evolve.

TEST METHODOLOGY

This report was produced by the RareDiseasePipeline Nextflow/Colab workflow, a synthetic-data validation harness modeled on production rare-disease WGS interpretation pipelines. Variants shown here were generated by the pipeline's own per-disease test-data generator (Section 12.1 of the pipeline notebook), then passed through the same region-filtering, annotation, disease-specific-branch, prioritization, and ACMG classification steps used for real inputs: VEP and SnpEff for functional consequence, ClinVar/SnpSift for clinical significance, gnomAD for population frequency, SpliceAI/REVEL/AlphaMissense/CADD for in-silico pathogenicity prediction, InterVar for ACMG/AMP classification, ClassifyCNV and SvAnna for structural-variant interpretation, ClinGen for gene dosage sensitivity, and ExpansionHunter/TRGT for repeat-expansion sizing. Variant classification follows the ACMG/AMP 2015 sequence-variant interpretation guidelines (Richards et al. 2015).

TEST LIMITATIONS

This is a synthetic dataset generated for pipeline validation and does not represent a real patient or real sequencing data. As with any WGS-based test, full genome coverage is not achievable in repetitive or duplicated regions, and certain variant classes (e.g. deep intronic variants, some repeat expansions, complex structural rearrangements) may not be reliably detected. Structural-variant and

inversion calls that are reported by only one tool in the chain are flagged as candidates and require orthogonal confirmation before they can be considered clinically actionable.

REGULATORY DISCLOSURES

This report was generated by a personal/educational bioinformatics pipeline (RareDiseasePipeline) built for pipeline development and validation purposes. It has not been developed, validated, or reviewed by any accredited clinical laboratory, has not been cleared or approved by the FDA, and is not a CLIA- or CAP-certified clinical test result. It must not be used for patient care, diagnosis, or any clinical decision-making.

LAB CONTACT

Email: pipeline-qa@example-lab.org (fictional, for template purposes only)

[Pipeline Owner Name]
Pipeline Developer / QA Lead

[Reviewer Name]
Secondary Reviewer (placeholder)

Test results reviewed and approved by: [Reviewer Name] — placeholder signature, synthetic report.

Patient	Case ID	DOB	Sex	Report Date	Page
SIM-DMD-01	CS-DMD_INXXXX	XX/XX/XXXX	Male	2026-07-15	1-3